Progressive Multi-Iteration Registration-Fusion Co-Optimization Network for Unregistered Hyperspectral Image Super-Resolution

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***Abstract*— Existing fusion-based hyperspectral image super-resolution (fusion-based HSI-SR) methods usually reconstruct high-resolution hyperspectral image (HR-HSI) by integrating the complementary information of low-resolution hyperspectral image (LR-HSI) and high-resolution multispectral image (HR-MSI). However, most of such methods rely on accurately registered images or consider registration and fusion as a two-stage task, which means that fusion must tolerate the accumulation of errors due to misregistration. In this article, we propose a progressive multi-iteration registration-fusion co-optimization network (PMI-RFCoNet) for unregistered HSI-SR, which progressively refines the registration and fusion result over multiple levels to reconstruct registered HR-HSI. To achieve registration-fusion co-optimization, the registration-fusion co-optimization block (Co-RFB) is designed to iterate continuously over multiple levels. We embed the interactive registration module (IRM) and the spectral recalibration and fusion module (SRFU) in Co-RFB, which can facilitate the network utilizing spatial and spectral features at different levels to generate more accurate HR-HSI. Specifically, IRM generates deformation field based on spatial correlations captured at long distances to repair non-rigid pixel offsets, and SRFU further performs adaptive high-fidelity spectral correction and spatial information fusion on the registration results. We conduct experimental verification on four widely used datasets, and the results show that PMI-RFCoNet can flexibly cope with different types and degrees of non-rigid deformation and achieve superior performance. Code is available at https://github.com/Jiahuiqu/PMI-RFCoNet.**

***Index Terms*— Co-optimization, hyperspectral image (HSI), non-rigid registration, super-resolution.**

NOMENCLATURE

Abbreviation	Description
HSI	Hyperspectral image.
HSI-SR	Hyperspectral image super-resolution.
HR-HSI	High-resolution hyperspectral image.
LR-HSI	Low-resolution hyperspectral image.
HR-MSI	High-resolution multispectral image.
PMI-RFCoNet	Progressive multi-iteration registration-fusion co-optimization network.
Co-RFB	Registration-fusion co-optimization block.
IRM	Interactive registration module.
SRFU	Spectral recalibration and fusion module.

I. INTRODUCTION

COMPARED with RGB images, the fine spectral information of HSI can more accurately reflect the characteristics of different ground objects [1], [2]. However, in the design of optical remote sensing systems, limited by equipment, incident energy, and other factors, there is a compromise between signal-to-noise ratio, spatial resolution, and spectral resolution, which makes it impossible for HSI to possess high spectral resolution and spatial resolution at the same time. The lack of spatial detail limits the application of HSI in fields such as ground object classification [3], [4], [5], [6], semantic segmentation [7], change detection [8], [9], [10], and anomaly detection [11], [12].

Super-resolution (SR) is an effective technique to recover detailed and natural spatial information from degraded low-resolution images, and has been widely studied in hyperspectral field. Existing HSI-SR methods based on deep learning are categorized into two groups based on whether auxiliary information is used or not: single HSI-SR and fusion-based HSI-SR [13], [14], [15]. Previous studies have shown that a single HSI-SR can obtain HR-HSI with high visual quality without any auxiliary images, which is more flexible in practical applications [16]. However, due to the extremely limited

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spatial resolution of HSI, it is often difficult for the network to recover accurate spatial detail information from the limited information, so fusion-based HSI-SR comes into being. Such methods fuse LR-HSI with HR-MSI, which makes the output image has more detailed information. In this article, we will focus on fusion-based HSI-SR.

Although fusion-based HSI-SR generally outperforms single HSI-SR, most of these methods only focus on fusion process and rely heavily on precise registration relationships between input images. However, in most cases, due to the influence of terrain undulation and atmospheric turbulence, it is impossible to realize strict spatial registration between images captured by different sensors, which greatly weakens the diversity of fusion-based HSI-SR in practical applications. In order to alleviate the strong assumption of fusion-based HSI-SR for accurate registration of input images, some recent works have started to consider the unregistration problem in the process of reconstructing HR-HSI. Zhou et al. [17], Ying et al. [18], and Fu et al. [19] regard registration and fusion as two-stage tasks. They first register by estimating the geometric transformation matrix of the input image pair, and then fuse the registered images. Although these methods improve the accuracy of HR-HSI reconstruction, they are only suitable for simple affine transformation, and it is difficult to deal with the more common non-rigid deformation in practice. Moreover, separating the registration and fusion tasks will force the fusion results to depend on the previous registration results, which is prone to error accumulation and poses an obstacle to reconstructing higher-quality HR-HSI. Qu et al. [20] and Zheng et al. [21] employ unsupervised framework for registration and fusion, but they both utilize the spectral information of LR-HSI to facilitate the network in generating HR-HSI based on HR-MSI, instead of solving the fundamental problem of registration.

Given the data characteristics of HSIs and the diversity of practical application scenarios, the existing research related to fusion-based HSI-SR for unregistered images still encounters the following challenges.

1) First, most of the existing fusion-based methods based on unregistered images still focus on simpler rigid transformations, and there is still less relevant work on joint optimization of non-rigid registration and fusion.
2) Second, images with large deformations often require the network to consider both global and local information. However, most of the existing methods often take a limited number of iterations, which not only ignore the utilization of global and local information, but also make it difficult to maintain their stability under complex deformation.
3) Third, most of the existing registration-fusion co-optimization algorithms ignore the correlation between the neighboring spectral bands, and it is easy to lose certain spectral features as the network deepens. In addition, pixel offsets caused by registration during the reconstruction process also bias some spectral features, which is not conducive to reconstructing more accurate HR-HSI.

In order to address the above problems, this article proposes a PMI-RFCoNet to reconstruct high-quality HR-HSI. As shown in Fig. 1, we skillfully integrate the registration and fusion into the Co-RFB and design it to be optimized iteratively at multiple levels. The Co-RFB at each level is able to accurately capture the correlations and differences between images over long distances, adaptively utilize the spectral prior information to recalibrate the characteristic responses among channels, and ensure that the PMI-RFCoNet achieves high-precision registration and fusion progressively. The contributions of this article are summarized as follows:

1) This article proposes an end-to-end PMI-RFCoNet. We model the registration and fusion into a framework and use LR-HSI and HR-MSI to perform registration and fusion in an iterative manner at multiple levels to progressively achieve HSI-SR.
2) We design the Co-RFB to be distributed at multiple levels for iterative optimization, which facilitates the network to refine the results step by step, and makes it possible to balance stability and accuracy when dealing with complex and large-scale image deformations.
3) Considering that effectively utilizing the inherent characteristics of HSI is crucial for HSI-SR, we design SRFU to adaptively explore the strong correlation between spectral features and exploit the spatial detail information on two branches, and realize the mutual promotion of spectral recalibration and spatial information fusion for feature maps.

The rest of the article is organized as follows. In Section II, we review some relevant works. The proposed PMI-RFCoNet is introduced in Section III. In Section IV, we present qualitative and quantitative results for four datasets with different deformations and perform the ablation studies. Conclusion is presented in Section V. The abbreviations used in this article and their definitions are shown in Nomenclature.

II. Related Work

Existing HSI-SR methods can generally be divided into two categories: single HSI-SR and fusion-based HSI-SR. In this section, we review some representative works to introduce their research status.

A. Single HSI-SR

Without auxiliary images, single HSI-SR has received considerable attention in reality. Some methods used a combination of CNN and traditional methods for algorithm optimization [22], [23]. However, most of these algorithms require intermediate intervention, which inevitably leads to error accumulation and lacks flexibility. Nowadays, the end-to-end SR framework has been proved that it can obtain better performance, and CNN and GAN are the most commonly adopted network structures. LN-atten-CNN [24] embedded both local and non-local 3-D attention modules in residual 3D-CNN, while focusing on local contextual information and remote interdependencies of HSI. Jiang et al. [16] conducted single HSI-SR based on progressive upsampling framework

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to alleviate the difficulty of feature extraction due to high-dimensional data. Liu et al. [25] unfolded the optimization process of the super-resolution prior-driven MAP framework as an interpretable multilevel network with a lightweight structure for resource-limited applications. With the application of transformer in the field of natural language processing [26], self-attention-based structures have also been studied. Interactformer [27] uses a two-branch architecture to capture remote dependencies and local features of HSI using transformer and 3-DCNN, respectively. Although the existing single HSI-SR methods have achieved good performance, the use of only a single LR-HSI as a network input is still limited by insufficient information.

B. Fusion-Based HSI-SR

Unlike single HSI-SR, fusion-based HSI-SR typically recovers the finer spatial information by fusing panchromatic image (PAN) or MSI with LR-HSI. Current fusion-based HSI-SR methods can be broadly classified into five categories: component substitution (CS), multiresolution analysis (MRA), Bayesian, matrix decomposition, and deep learning. The CS methods [28] replace the spatial information obtained by HSI separation in other domains with MSI, and then obtain the fused image by inverse transformation. The MRA methods [29], [30] inject spatial information extracted from MSI into HSI at different scales. Bayesian [31] uses prior constraints to maximize the posterior probability density of the HSI-SR task. The core idea of matrix factorization is spectral unmixing and matrix optimization, its representative algorithm is coupled nonnegative matrix factorization (CNMF) [32].

In recent years, researchers have proposed many fusion-based HSI-SR methods based on deep learning. Dong et al. [33] unfolded the algorithm based on a deep HSI denoiser into a novel model-guided deep convolutional network with excellent performance. Li et al. [34] proposed a two-stage learning methods with excellent results, which improves the image-specific details by multistep propagation in the coarse stage with cyclic learning of image-specific details in the fine stage. MulSR [35] explores the intrinsic relationship between different resolutions and utilized small-scale factor models to assist models of large-scale factors to achieve joint SR. The above methods are all based on supervised learning, although they have achieved good performance, it is difficult to apply these methods in reality when there is no HR-HSI for training. Qu et al. [36] made the first attempt to solve the HSI-SR problem in unsupervised manner. The proposed uSDN makes the HSI and MSI follow the sparse Dirichlet distribution to naturally contain the physical constraints. Li et al. [37] innovatively performing band-by-band SR in the coarse stage and proposing a plug-and-play unsupervised method to refine the coarse results in the fine stage, which achieves significant gains in both spatial reconstruction and spectral fidelity. The IUPNet proposed by Qu et al. [38] is a novel interpretable unsupervised network based on observation model, which contains a generator that uses two discriminators to build an adversarial game in each iteration to update the information. Guo et al. [39] explored cross-domain consistency differences and dual domain high-frequency details in both spatial and frequency domains, which effectively improved the generalization ability of unsupervised learning.

All of the above methods assume a well-registered relationship between LR-HSI and HR-MSI, and recent studies have attempted to design new models to deal with unregistered data. Ying et al. [18] introduced normalized edge difference (NED) for image similarity measure in the registration stage, and added interpolation to the spatial degradation model in the fusion stage. This method is a two-stage task, which may inevitably result in error accumulation. u^2-MDN [20] characterizes spatial correlation between different modes by maximizing the mutual information. Zheng et al. [21] proposed NonRegSRNet which facilitates registration and fusion by learning spatial and spectral response functions. These two methods utilize the spectral information of LR-HSI in order to facilitate the network to reconstruct HR-HSI based on HR-MSI, which is essentially a mutual information-based idea.

III. Proposed Method

In this section, we will introduce the method in detail. As shown in Fig. 1, the Co-RFBs over multiple layers are the core of PMI-RFCoNet. Co-RFB mainly consists of two parts: IRM, generating deformation field based on spatial inter-correlation captured at long distances to repair pixel offsets, and SRFU, fully exploiting spectral correlation and spatial details to realize spectral recalibration and spatial information fusion. In the following, we will first formulate the problem, then introduce the network framework, and finally detail the components of Co-RFB.

A. Problem Formulation

In this article, the problem we are working on is how to achieve fusion-based HSI-SR using unregistered data sources with non-rigid deformation. It is noted that tensors and matrices are denoted by bold capital letters, and vectors are denoted by bold lowercase letters. First, we assume that HR-MSI is the fixed image and LR-HSI is the moving image. Then, let $\mathbf{M}_f \in \mathbb{R}^{H\times W\times b}$ denotes HR-MSI, H and W represent the height and width of $\mathbf{M}_f$, and b represents the number of spectral bands of $\mathbf{M}_f$. $\mathbf{H}_m \in \mathbb{R}^{h\times w\times B}$ denotes LR-HSI, h and w represent the height and width of $\mathbf{H}_m$, and B represents the number of spectral bands of $\mathbf{H}_m$. In this article, our purpose is to generate HR-HSI $\mathbf{X}_f \in \mathbb{R}^{H\times W\times B}$ using unregistered $\mathbf{M}_f$ and $\mathbf{H}_m$, and $\mathbf{X}_f$ is spatially consistent with $\mathbf{M}_f$. According to the Wald's protocol [40], $\mathbf{M}_f$ and $\mathbf{H}_m$ can be considered as degraded images of $\mathbf{X}_f$ in spectral and spatial dimensions, respectively. In addition, due to many factors during the imaging process, the acquired $\mathbf{H}_m$ often has geometric distortion and cannot maintain a completely consistent spatial correspondence with $\mathbf{M}_f$. The relationship between $\mathbf{X}_f$ and acquired $\mathbf{M}_f$ and $\mathbf{H}_m$ can be expressed as

$$\mathbf{M}_f = \mathbf{X}_f\mathbf{R} \tag{1}$$

$$\mathbf{H}_m = f_d(\mathbf{S}\mathbf{X}_f) \tag{2}$$

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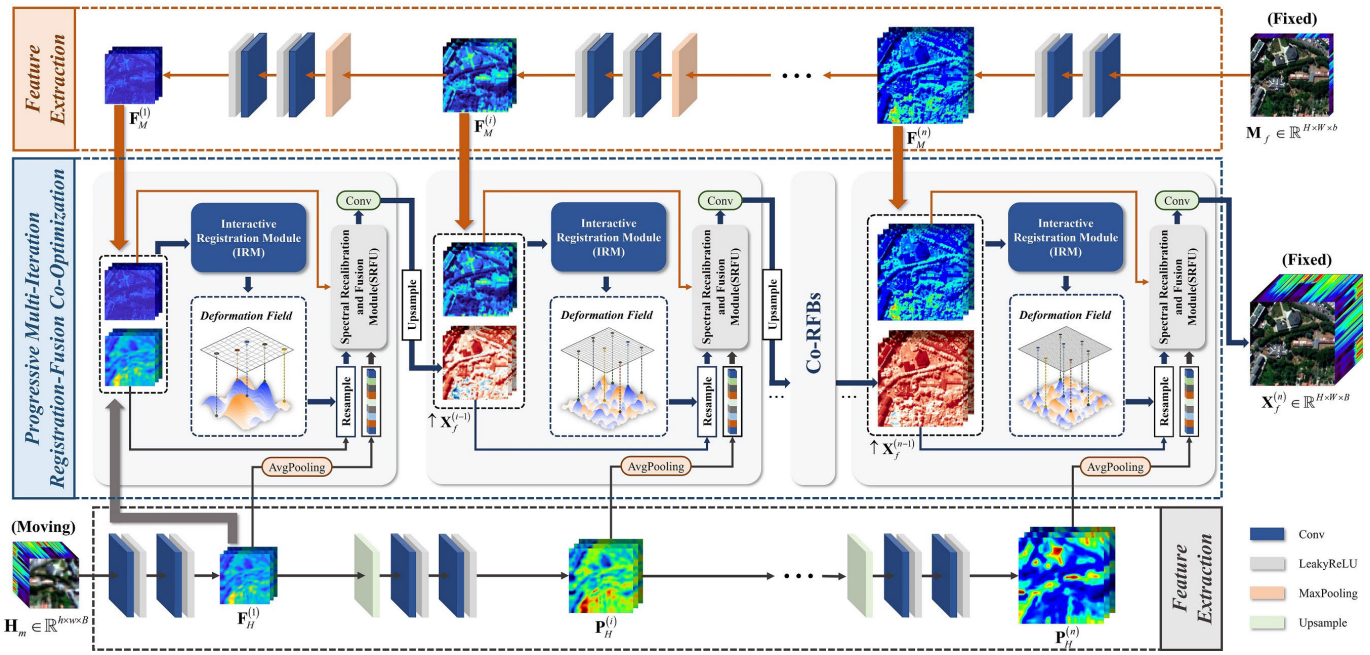


Fig. 1. Architecture of the proposed PMI-RFCoNet for HSI-SR, which utilizes unregistered LR-HSI $\mathbf{H}_m$ and HR-MSI $\mathbf{M}_f$ to accurately reconstruct registered HR-HSI $\mathbf{X}_f^{(n)}$ from global to local. Spatial and spectral features under different levels are fed into the iterative Co-RFBs, allowing the network to achieve progressive multi-iteration registration-fusion co-optimization.

where $\mathbf{S} \in \mathbb{R}^{HW \times hw}$ represents the spatial transformation matrix based on the spatial blurring kernel and downsampling, $\mathbf{R} \in \mathbb{R}^{B \times b}$ represents the spectral response matrix based on the spectral blur kernel, and $f_d(\cdot)$ represents deformation function which causes geometrical distortion in LR-HSI $\mathbf{H}_m$.

As described in the related work, current deep learning methods for solving HSI-SR based on unregistered images often process registration and fusion separately, it can be expressed as follows:

$$\mathbf{X}_f = \mathcal{F}_\theta\big(\mathcal{D}\big(\mathbf{M}_f, \mathbf{H}_m\big), \mathbf{M}_f\big) \tag{3}$$

where $\mathcal{D}(\cdot)$ represents the registration method, and $\mathcal{F}_\theta(\cdot)$ represents the fusion network parameterized by θ. This method regards registration and fusion as two independent steps, and the errors generated during the registration process will accumulate to the subsequent fusion step. Registration-fusion co-optimization integrates the two steps into the same model to achieve the complementarity and improve the final fusion performance.

In this article, we aim to further improve (3). We hope that the improved network can adaptively learn the differences and connections between images with complex deformation, and enhance the reconstruction effect by adjusting the local details more finely while maintaining the global consistency. Therefore, we integrate registration and fusion into one block, making it possible to utilize LR-HSI and HR-MSI features at multiple levels for progressive iteration to optimize the results. We can formulate as follows:

$$\mathbf{X}_f^{(i)} = \mathcal{T}_\theta^{(i)}\left(\left[\mathcal{F}\big(\mathbf{M}_f, \mathbf{H}_m\big)\right]^{(i)}, \mathbf{X}_f^{(i-1)}\right) \tag{4}$$

where i represents the number of levels, $\mathcal{F}(\cdot)$ represents the feature extraction of $\mathbf{M}_f$ and $\mathbf{H}_m$ at the ith level, $\mathcal{T}_\theta^{(i)}(\cdot)$ represents the Co-RFB with θ as the parameter at the ith level, $\mathbf{X}_f^{(i-1)}$ can be considered as the rough result at the lower resolution level, and $\mathbf{X}_f^{(i)}$ is the finer result at the higher resolution level.

B. Network Architecture

In this section, we will describe the general framework of the proposed PMI-RFCoNet. As shown in Fig. 1, $\mathbf{H}_m$ and $\mathbf{M}_f$, respectively, denote the acquired LR-HSI and HR-MSI, and assuming that HR-MSI is the fixed image and LR-HSI is the moving image with deformation.

First, we design two feature extractors to extract texture and spectral features from HR-MSI and LR-HSI step by step, respectively. The spatial texture features obtained from HR-MSI are represented by $\{\mathbf{F}_M^{(n)}, \ldots, \mathbf{F}_M^{(i+1)}, \mathbf{F}_M^{(i)}, \ldots, \mathbf{F}_M^{(1)}\}$. Their spatial resolution is gradually transitioned from the finest level n to the coarsest level 1, where the feature map size of the ith level is $1/2$ of the $(i+1)$th level. Formally, we define

$$\mathbf{F}_M^{(i)} = \begin{cases} f_{\text{conv},3}^{j}\big(\mathbf{M}_f\big), & i = n \\ f_{\text{conv},3}^{j}\Big(\text{AvgPooling}\Big[\mathbf{F}_H^{(i+1)}\Big]\Big), & 1 \le i < n \end{cases} \tag{5}$$

where $f_{\text{conv},3}^{j}(\cdot)$ represents the combination of 3×3 convolutional layers and LeakyReLU layers, j represents the number of the above combinations, and AvgPooling$(\cdot)$ represents global average pooling. The features obtained from LR-HSI are represented by $\{\mathbf{F}_H^{(1)}, \mathbf{P}_H^{(1)}, \ldots, \mathbf{P}_H^{(i)}, \mathbf{P}_H^{(i+1)}, \ldots, \mathbf{P}_H^{(n)}\}$, where $\mathbf{F}_H^{(1)}$ represents spatial texture feature and the rest represent spectral features. Their spatial resolution is gradually transitioned from the coarsest level 1 to the finest level n, where the feature map size of the $(i+1)$th level is twice of

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the ith level. Formally, we define

$$\mathbf{F}_H^{(1)} = f_{\text{conv},3}^{j}(\mathbf{H}_m) \tag{6}$$

$$\mathbf{P}_H^{(i)} = \begin{cases} f_{\text{conv},1}^{j}(\mathbf{H}_m), & i = 1 \\ f_{\text{conv},1}^{j}\left(\text{TransConv}\left[\mathbf{P}_H^{(i-1)}\right]\right), & 1 < i \leq n \end{cases} \tag{7}$$

where $f_{\text{conv},1}^{j}(\cdot)$ represents the combination of 1×1 convolutional layers and LeakyReLU layers, and TransConv$(\cdot)$ represents transposed convolutional layer. Then, we design Co-RFB distributed in multiple levels. The output of Co-RFB in each layer will gradually transition from low resolution to high resolution. We consider the output of the $(i-1)$th Co-RFB as the rough result, and the ith $(i \neq 1)$ Co-RFB will utilize the rough result, $\mathbf{F}_M^{(i)}$, and $\mathbf{P}_H^{(i)}$ to achieve further optimization. We can express the above process as follows:

$$\mathbf{X}_f^{(i)} = \begin{cases} \mathcal{T}_\theta^{(i)}\left(\mathbf{F}_H^{(i)}, \mathbf{F}_M^{(i)}, \mathbf{P}_H^{(i)}\right), & i = 1 \\ \mathcal{T}_\theta^{(i)}\left(\uparrow \mathbf{X}_f^{(i-1)}, \mathbf{F}_M^{(i)}, \mathbf{P}_H^{(i)}\right), & 1 < i \leq n \end{cases} \tag{8}$$

where $\mathcal{T}_\theta^{(i)}(\cdot)$ represents the Co-RFB at the ith level, $\uparrow \mathbf{X}_f^{(i-1)}$ and $\mathbf{X}_f^{(i)}$ represent the input and output at the ith level, respectively. Section III-C provide a detailed description of the proposed Co-RFB.

C. Registration-Fusion Co-Optimization Block

As mentioned in the previous section, we integrate registration and fusion into Co-RFB, which motivates the network to reconstruct HR-HSI progressively. For the ith level Co-RFB, the two basic components are the IRM and the SRFU. The output of Co-RFB at the ith level can be expressed as

$$\mathbf{X}_f^{(i)} = \mathcal{M}_S\left(\mathbf{X}_{\text{reg}}^{(i)}, \mathbf{F}_M^{(i)}, \mathbf{P}_H^{(i)}\right) \tag{9}$$

where $\mathcal{M}_S(\cdot)$ is the function of the proposed SRFU, $\mathbf{F}_M^{(i)}$ and $\mathbf{P}_H^{(i)}$ represent the spatial and spectral features at the ith level, and $\mathbf{X}_{\text{reg}}^{(i)}$ denotes the registration result at the ith level. $\mathbf{X}_{\text{reg}}^{(i)}$ is obtained by resampling the output $\uparrow \mathbf{X}_f^{(i-1)}$ of the $(i-1)$th Co-RFB using the deformation field generated by IRM. The above resampling is an assignment process. Specifically, for each pixel $\mathbf{p}$ of $\mathbf{X}_{\text{reg}}^{(i)}$, the corresponding pixel $\mathbf{p}' = \mathbf{p} + \mathbf{U}(\mathbf{p})$ can be found in $\uparrow \mathbf{X}_f^{(i-1)}$, where $\mathbf{U}(\mathbf{p})$ represents the displacement. It should be noted that $\mathbf{p}'$ may be a decimal, but $\uparrow \mathbf{X}_f^{(i-1)}$ only has values where the pixel coordinates are integers, so we choose to perform bilinear interpolation on the four pixel values with integer coordinates around $\mathbf{p}'$ to obtain the value of $\mathbf{p}'$. After that, we can complete the resampling of $\uparrow \mathbf{X}_f^{(i-1)}$ by assigning the value of $\mathbf{p}'$ to $\mathbf{p}$ and obtain $\mathbf{X}_{\text{reg}}^{(i)}$. The above procedures can be expressed as follows:

$$\begin{aligned} \mathbf{X}_{\text{reg}}^{(i)}(\mathbf{p}) &= \uparrow \mathbf{X}_f^{(i-1)}(\mathbf{p}') \\ &= \uparrow \mathbf{X}_f^{(i-1)} \circ \phi(\mathbf{p}) \\ &= \sum_{\mathbf{q} \in \mathcal{Z}(\mathbf{p}')} \uparrow \mathbf{X}_f^{(i-1)}(\mathbf{q}) \prod_{d \in \{x,y\}} \left(1 - \left|\mathbf{p}'_d - \mathbf{q}_d\right|\right) \end{aligned} \tag{10}$$

$$\phi = \mathcal{M}_C\left(\mathbf{F}_M^{(i)}, \uparrow \mathbf{X}_f^{(i-1)}\right) \tag{11}$$

where $\uparrow \mathbf{X}_f^{(i-1)}$ denotes the result after upsampling the output of the $(i-1)$th Co-RFB, $\mathcal{Z}(\mathbf{p}')$ are the neighbors of $\mathbf{p}'$, $\mathcal{M}_C(\cdot)$ is the function of the proposed IRM, and ϕ denotes the deformation field. The proposed IRM and SRFU are described in detail as follows.

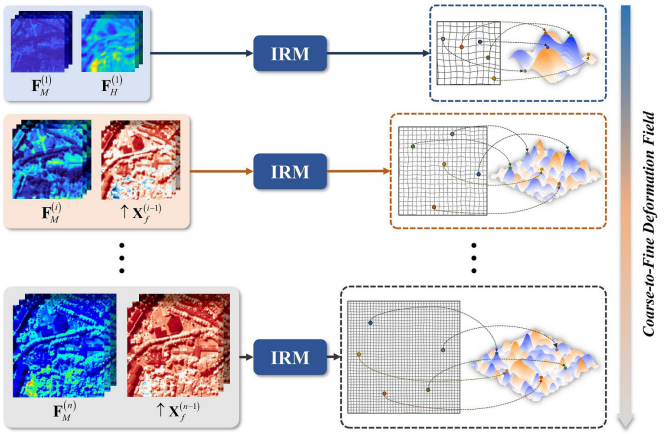


Fig. 2. Coarse-to-fine deformation field generated by IRM at different levels. IRM focuses on overall deformation trend at lower resolution levels and makes finer adjustments to local details at higher resolution levels.

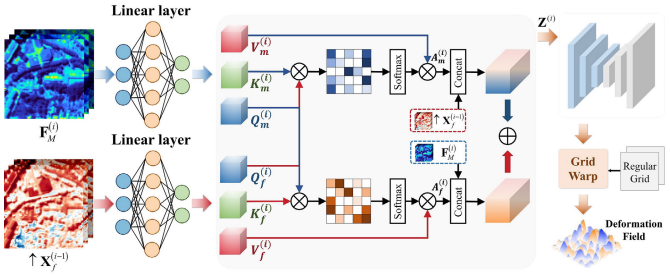


Fig. 3. Architecture of the IRM, which constantly focuses on the correlations and differences between input features based on information interaction to generate deformation fields toward non-rigid registration.

1) Interactive Registration Module: Unlike rigid registration, non-rigid registration aims to find a nonlinear transformation function that maps each pixel in an image to a corresponding position in another image. When the input image undergoes more complex non-rigid deformations, the inaccuracy of the registration can directly lead to unwanted artifacts or distortions. In order to reduce the negative impact of such spatial dislocations on the subsequent reconstruction, we introduce IRM to globally capture correlation information between regions and generate deformation field to repair non-rigid registration. As shown in Fig. 2, the deformation field reflects the overall deformation trend at lower resolution levels and focuses on local details at higher resolution levels.

In the following, we will describe the IRM in detail. The input feature maps are $\mathbf{F}_M^{(i)}$ and $\uparrow \mathbf{X}_f^{(i-1)}$, where $\mathbf{F}_M^{(i)}$ denotes the features extracted by HR-MSI under the ith level and $\uparrow \mathbf{X}_f^{(i-1)}$ denotes the output of the previous Co-RFB. As shown in Fig. 3, we first use linear layers for mapping $\mathbf{F}_M^{(i)}$ and $\uparrow \mathbf{X}_f^{(i-1)}$ to obtain two different sets of query vectors $\mathbf{Q}$ (Query), key vectors $\mathbf{K}$ (Key), and $\mathbf{V}$ (Value)

$$\mathbf{Q}_m^{(i)}, \mathbf{K}_m^{(i)}, \mathbf{V}_m^{(i)} = f_m^i\left(\mathbf{F}_M^{(i)}\right) \tag{12}$$

$$\mathbf{Q}_f^{(i)}, \mathbf{K}_f^{(i)}, \mathbf{V}_f^{(i)} = f_f^i\left(\uparrow \mathbf{X}_f^{(i-1)}\right) \tag{13}$$

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where $f_m^i(\cdot)$ represents the mapping of $\mathbf{F}_M^{(i)}$ to obtain $\mathbf{Q}_m^{(i)}$, $\mathbf{K}_m^{(i)}$ and $\mathbf{V}_m^{(i)}$, and $f_f^i(\cdot)$ represents the mapping of $\uparrow \mathbf{X}_f^{(i-1)}$ to obtain $\mathbf{Q}_f^{(i)}$, $\mathbf{K}_f^{(i)}$, and $\mathbf{V}_f^{(i)}$. In order to realize the information interaction, we next use the form of crossover to generate feature representations containing spatial correlations based on the above vectors on two branches, which can reflect the most globally correlated positions of the input feature maps. Then, we obtain two spatially correlated attention maps by multiplying the two inter-correlation matrices with the corresponding value vectors

$$\mathbf{A}_m^{(i)} = \mathrm{MatMul}\left(\mathrm{Softmax}\left(\frac{\mathbf{Q}_f^{(i)}\left[\mathbf{K}_m^{(i)}\right]^{\mathrm{T}}}{\sqrt{C}}\right), \mathbf{V}_m^{(i)}\right) \tag{14}$$

$$\mathbf{A}_f^{(i)} = \mathrm{MatMul}\left(\mathrm{Softmax}\left(\frac{\mathbf{Q}_m^{(i)}\left[\mathbf{K}_f^{(i)}\right]^{\mathrm{T}}}{\sqrt{C}}\right), \mathbf{V}_f^{(i)}\right) \tag{15}$$

where $\mathbf{A}_m^{(i)}$ reflects the importance of $\mathbf{F}_M^{(i)}$ at the corresponding position of $\uparrow \mathbf{X}_f^{(i-1)}$, and $\mathbf{A}_f^{(i)}$ reflects the importance of $\uparrow \mathbf{X}_f^{(i-1)}$ at the corresponding position of $\mathbf{F}_M^{(i)}$. $\mathbf{A}_m^{(i)}$ and $\mathbf{A}_f^{(i)}$ help to focus on the difference and correlation information in the image during the subsequent generation of the deformation field. Finally, since the generation of the deformation field needs to ensure that the semantic information is consistent with the spatial correlation information in terms of location, $\mathbf{A}_m^{(i)}$ and $\mathbf{A}_f^{(i)}$ need to be connected with their corresponding $\uparrow \mathbf{X}_f^{(i-1)}$ and $\mathbf{F}_M^{(i)}$, and the obtained $\mathbf{Z}^{(i)}$ will have a richer feature representation. The above process can be formulated as follows:

$$\mathbf{Z}^{(i)} = \mathrm{Add}\left(\mathrm{Concat}\left(\mathbf{A}_m^{(i)}, \uparrow \mathbf{X}_f^{(i-1)}\right), \mathrm{Concat}\left(\mathbf{F}_M^{(i)}, \mathbf{A}_f^{(i)}\right)\right) \tag{16}$$

where $\mathbf{Z}^{(i)}$ contains both input feature information and introduces spatially correlated attention maps at their corresponding locations. This will motivate the subsequent generation of spatial transformation parameters to be more focused on certain regions which reflect the correlations between the input features.

Next, we feed $\mathbf{Z}^{(i)}$ into an encoder-decoder with jump connections to generate spatial transformation parameters. The final output spatial transformation parameters $\mathbf{U}$ is a matrix of size $H \times W \times 2$. For each pixel $\mathbf{p}$, $\mathbf{U}(\mathbf{p})$ is a displacement such that $\mathbf{X}_{\mathrm{reg}}^{(i)}(\mathbf{p})$ and $[\uparrow \mathbf{X}_f^{(i-1)} \circ \phi(\mathbf{p})]$ correspond to similar positions, where the deformation field ϕ is generated by warping the standard grid, denoted as

$$\phi = \mathrm{Id} + \mathbf{U} \tag{17}$$

where Id denotes the identity transform that describes the coordinate positions of the pixels.

2) Spectral Recalibration and Fusion Module: HSI usually has hundreds of consecutive spectral bands with strong correlations. Most of the existing registration-fusion co-optimization algorithms ignore the preservation of spectral features during the process. As the number of network layers increases, some subtle spectral features and inter-spectral correlations are easily blurred or lost. In addition, some spectral features may be biased due to pixel offsets from registration, which makes an obstacle in reconstructing high-quality HR-HSI. To alleviate the above problems, we design the SRFU as shown in Fig. 4. It performs spectral recalibration and spatial information fusion in two independent branches respectively, and there is some interaction between the two branches, which strengthens and promotes these two tasks alternately.

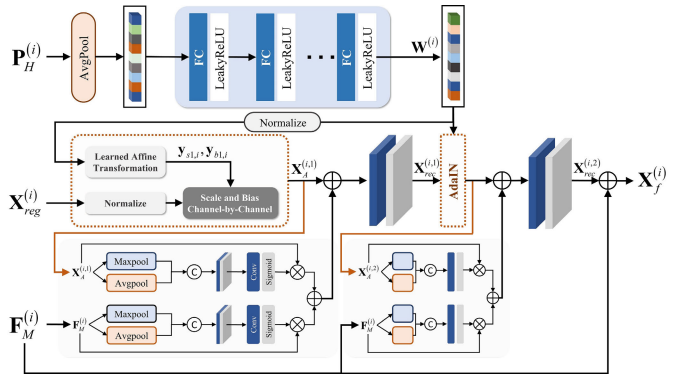


Fig. 4. Architecture of the SRFU. It performs spectral correlation embedding and spatial information fusion on the output of IRM interactively on two branches, which achieves adaptive high-fidelity spectral correction and detail enhancement.

In order to learn the dependencies between spectra, we first use an adaptive average pooling layer to transform the spectral features $\mathbf{P}_H^{(i)}$ at the ith level into a 1-D vector, which can be considered as a representation of global spectral features. Next, we apply fully connected layers to learn the nonlinear interactions between the 1-D vector to further resolve the dependencies between the spectra. The above process can be formulated as follows:

$$\mathbf{W}^{(i)} = f_{\mathrm{FC}}\left(\mathrm{AdaPooling}\left(\mathbf{P}_H^{(i)}\right)\right) \tag{18}$$

where $\mathbf{W}^{(i)}$ represents the 1-D feature representation obtained after fully connected layers, $f_{\mathrm{FC}}(\cdot)$ represents the fully connected layer, and AdaPooling$(\cdot)$ represents the adaptive average pooling layer. To achieve spectral recalibration using $\mathbf{W}^{(i)}$, we use adaptive instance normalization (AdaIN) to globally adjust the feature map channel-by-channel. We first normalize $\mathbf{W}^{(i)}$ and obtain the scaling and translation coefficients after a learnable affine transformation. Next, we normalize $\mathbf{P}_H^{(i)}$, and then use the obtained coefficients to recalibrate $\mathbf{X}_{\mathrm{reg}}^{(i)}$ channel-by-channel. We can define the process as follows:

$$\mathbf{y}_{s1,i}, \mathbf{y}_{b1,i} = f_{\mathrm{affine}}\left(\mathrm{Norm}\left(\mathbf{W}^{(i)}\right)\right) \tag{19}$$

$$\begin{aligned}\mathbf{X}_A^{(i,1)} &= f_{\mathrm{AdaIN}}\left(\mathbf{X}_{\mathrm{reg}}^{(i)}, \mathbf{y}_{s1,i}, \mathbf{y}_{b1,i}\right) \\ &= \mathbf{y}_{s1,i}\frac{\mathbf{X}_{\mathrm{reg}}^{(i)} - \mu\left(\mathbf{X}_{\mathrm{reg}}^{(i)}\right)}{\sigma\left(\mathbf{X}_{\mathrm{reg}}^{(i)}\right)} + \mathbf{y}_{b1,i}\end{aligned} \tag{20}$$

where $\mathbf{y}_{s1,i}$ and $\mathbf{y}_{b1,i}$, respectively, represent the learnable scaling and translation coefficients, $\mathbf{X}_A^{(i,1)}$ represents the output features after the AdaIN layer, $f_{\mathrm{affine}}(\cdot)$ represents the learnable affine transformation, Norm$(\cdot)$ represents regularization, $\mu(\mathbf{X}_{\mathrm{reg}}^{(i)})$ and $\sigma(\mathbf{X}_{\mathrm{reg}}^{(i)})$ represent the mean and variance of $\mathbf{X}_{\mathrm{reg}}^{(i)}$. Then, we send $\mathbf{X}_A^{(i,1)}$ and $\mathbf{F}_M^{(i)}$ into the spatial information

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TABLE I
DETAILED INFORMATION OF PUBLIC DATASETS AND REAL-WORLD DATASET

Dataset	Sensor	Range(nm)	Original size	Used size	Size of LR-HSI	Size of HR-MSI
Pavia Center	ROSIS	430 - 960	1096 × 1096 × 115	960 × 640 × 102	40 × 40 × 102	160 × 160 × 4
Houston	ITRES CASI-1500	380 - 1050	1905 × 349 × 144	1280 × 320 × 144	40 × 40 × 144	160 × 160 × 4
Chikusei	Headwall Hyperspec-VNIR-C	363 - 1018	2304 × 2304 × 128	2304 × 2048 × 128	64 × 64 × 128	256 × 256 × 4
WHU-OHS	OHS	400 - 1000	512 × 512 × 32	512 × 512 × 32	128 × 128 × 32	512 × 512 × 4
XijingU (HSI)	Co-Aligned HP	400 - 2500	960 × 960 × 137	960 × 960 × 137	160 × 160 × 137	-
XijingU (MSI)	FLIR Duo Pro R	-	3840 × 3840 × 3	3840 × 3840 × 3	-	640 × 640 × 3

fusion branch together. We generate spatial attention weights for both using maximum pooling layer, average pooling layer, and convolutional layer, respectively, and use the weights to weight the feature maps to extract more useful detail information. The two weighting results are then added together and fed into the spectral correction branch to achieve further information integration through the subsequent convolutional and activation function layers. Thus, we define

$$\mathbf{X}_{\text{rec}}^{(i,1)} = f_{\text{conv},3}\left(\text{Add}\left(f_{\text{spa}}\left(\mathbf{X}_A^{(i,1)}, \mathbf{F}_M^{(i)}\right), \mathbf{X}_A^{(i,1)}\right)\right). \tag{21}$$

It should be noted that we alternated the combination of AdaIN layer, convolutional layer, and LeakyReLU layer twice to get the final output of the SRFU

$$\mathbf{X}_{\text{rec}}^{(i,2)} = f_{\text{conv},3}\left(\text{Add}\left(f_{\text{spa}}\left(\mathbf{X}_A^{(i,2)}, \mathbf{F}_M^{(i)}\right), \mathbf{X}_A^{(i,2)}\right)\right) \tag{22}$$

$$\mathbf{X}_f^{(i)} = \text{Add}\left(\mathbf{F}_M^{(i)}, \mathbf{X}_{\text{rec}}^{(i,2)}\right) \tag{23}$$

where $f_{\text{rec}}(\cdot)$ represents the combination of AdaIN layer, convolutional layer, and LeakyReLU layer.

IV. EXPERIMENT RESULTS AND ANALYSIS

In this section, we validate the performance of the proposed method under different experimental conditions. First, we present the used public datasets, experimental setup, and evaluation indicators. Then, we compare PMI-RFCoNet with existing fusion-based HSI-SR methods from subjective and objective evaluation perspectives. Finally, we perform ablation studies to validate the effectiveness of the method.

A. Datasets and Evaluation Indicators

We validate our method on four public HSI datasets and one real-world HSI dataset. The four public datasets are Pavia Center [41], Houston [42], Chikusei [43], and WHU-OHS [44]. The details of the above datasets are shown in Table I.

1) Pavia Center: The Pavia Center dataset with size 960 × 640 × 102 is divided into 12 non-overlapping patches, nine patches are randomly selected as the training set, and the remaining three are used as the testing set. In order to expand the sample, each non-overlapping patch is slid to generate 21 patches of size 160 × 160 × 102 with a fixed stride.

2) Houston: The Houston dataset with size 1280 × 320 × 144 is divided into eight non-overlapping patches, of which six patches are used for training and the remaining two for testing. Each non-overlapping patch is divided into 41 patches of size 160 × 160 × 144 with a fixed step size.

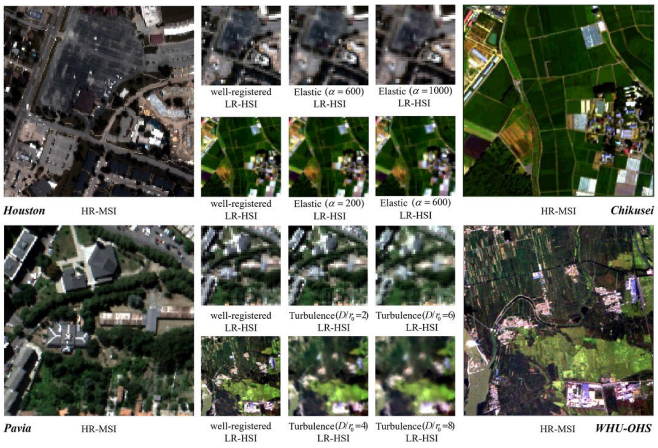


Fig. 5. Visualization of four public datasets simulating different degrees of elastic and turbulence deformation.

3) Chikusei: The Chikusei dataset with size 2304 × 2048 × 128 is divided into 36 non-overlapping patches, of which 27 patches are used as the training set and the remaining nine as the testing set. Each patch is further divided into eight patches of size 256 × 256 × 128 in fixed steps to expand the samples.

4) WHU-OHS: The WHU-OHS dataset covers 42 regions in China, we selected Jilin for our experiments, and the original patch sizes are all 512 × 512 × 32. After removing the abnormal data and enlarging the training samples, there are finally 334 patches with size 256 × 256 × 32 used for training and 23 patches with size 512 × 512 × 32 used for testing.

5) XijingU: The XijingU dataset was collected by our team over Xijing University in Xi'an, Shaanxi, China, named XijingU dataset. This dataset contains an LR-HSI of size 960 × 960 × 137 and an HR-MSI of size 3840 × 3840 × 3.

6) Dataset Processing Details: For the four public datasets, we take the acquired HSI patch as the reference and generate LR-HSI and HR-MSI according to Wald's protocol. For LR-HSI, we first use an isotropic Gaussian blurring kernel to blur the reference image, then downsample it to 1/4 of its original spatial size. HR-MSI is obtained by using the blue-green-red-near-infrared spectral response function on the reference image.

For the XijingU dataset, due to the unavailability of reference images, we construct the training set by using Wald's protocol to obtain LR-HSI of size 240 × 240 × 137 and HR-MSI of size 960 × 960 × 3. Then, we expand the training

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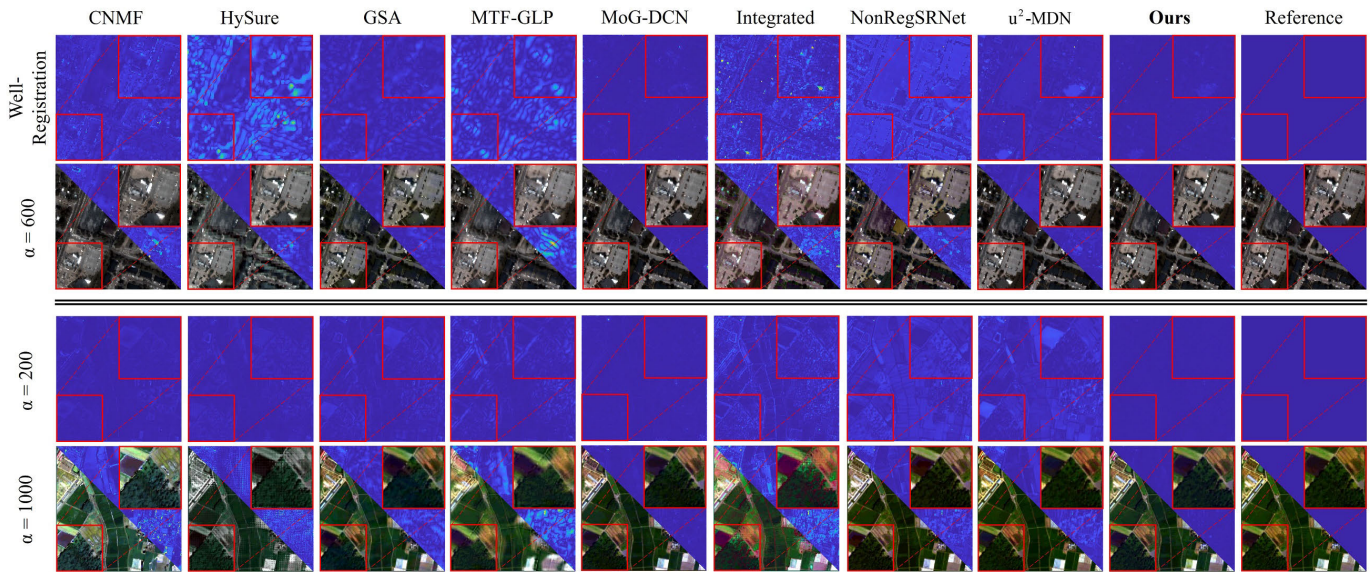


Fig. 6. Visualization results of different methods on Houston and Chikusei datasets with different degrees of elastic deformation.

samples by overcropping the samples to obtain 69 pairs of images with sizes of 40 × 40 × 137 and 160 × 160 × 3, respectively. For testing, the real images are cropped to obtain 69 pairs of LR-HSI with size 160 × 160 × 137 and HR-MSI with size 640 × 640 × 3.

7) Evaluation Indicators: For the four public datasets, we select six widely used spatial and spectral evaluation metrics for quantitative assessment, which are peak signal-to-noise ratio (PSNR), cross-correction (CC), spectral angle mapping (SAM), errur relative globale adimensionnelle desynthese (ERGAS), root-mean-square error (RMSE), and structure similarity index measure (SSIM). The higher the PSNR, CC, and SSIM, and the lower the SAM, ERGAS, and RMSE, the higher the quality of the fused image. These metrics are widely used in the field of HSI fusion processing, which is helpful for the comprehensive evaluation of the model from the objective level.

For the XijingU dataset, we select the quality without reference (QNR) and two derived indexes (D_λ and D_s) that quantify the spectral and spatial distortions as the basis for evaluation. The higher the QNR index, the higher the quality of the obtained HR-HSI. The maximum theoretical value of this index is 1 when both D_λ and D_s are equal to 0.

B. Experimental Setup Details

1) Simulated Details: For obtaining the unregistered image pairs, we employ elastic transformation [45] and turbulence transformation [46] for LR-HSI to simulate the non-rigid deformations that may occur during the image acquisition process in reality.

In order to more effectively evaluate the model's adaptability to slight and extreme deformations as well as to assess its generalization capability, we also set different deformation degrees from small to large. As shown in Fig. 5, we simulate different degrees of elastic deformation for the Houston and Chikusei datasets. α is set to 200, 600, and 1000, which represents a control factor that determines the strength of deformation. For the Pavia Center and WHU-OHS datasets, we simulate different degrees of turbulence deformation for them. The parameter $\lambda = (D/r_0)$ is set to 2, 4, 6, and 8. The larger λ is, the greater degree of blurring and tilting, and the stronger influence by turbulence.

2) Loss Function: In our experiments, we choose L_1 as the loss function to train the proposed PMI-RFCoNet. L_1 loss function is considered as the preferred choice for the fusion-based HSI-SR task due to its advantage in preserving the image edge and texture structure, as well as its ability to efficiently penalize small absolute errors to ensure better convergence in the training phase. Specifically, the L_1 loss function is defined as

$$\mathcal{L}(\Theta) = \frac{1}{N}\sum_{n=1}^{N}\left\|\mathbf{X}_r^n - f_{\text{Net}}\left(\mathbf{H}_m^n, \mathbf{M}_f^n\right)\right\|_1 \tag{24}$$

where $f_{\text{Net}}(\mathbf{H}_m^n, \mathbf{M}_f^n)$ and $\mathbf{X}_r^n$ are the nth reconstructed HR-HSI and the reference HSI, respectively, N denotes the number of images in one training batch, and Θ represents the parameter set of the proposed PMI-RFCoNet.

3) Experimental Environment: The proposed network is implemented based on the Pytorch framework and trained on NVIDIA GTX 3090 GPUs. In the training phase, we update the parameters using the Adam optimizer. We set the batch size to 4, the total number of epochs to 400, and the learning rate is initialized to 0.001 and decayed by a factor of 10 at epoch = 50, 150, and 300, respectively.

C. Performance Comparison

In order to fully validate the effectiveness of PMI-RFCoNet in this article, we compare it with other methods: CNMF [32], HySure [31], GSA [28], MTF-GLP [30], MoG-DCN [33], Integrated [17], NonRegSRNet [21], and u^2-MDN [20]. Among them, CNMF, HySure, GSA, and MTF-GLP are model-based traditional methods, and the rest are DL-based

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TABLE II

QUANTITATIVE COMPARISON ON HOUSTON DATASET WITH VARYING DEGREES OF ELASTIC DEFORMATION. (COLOR CONVENTION: **BEST** AND SECOND-BEST)

Method		CNMF	HySure	GSA	MTF-GLP	MoG-DCN	Integrated	NonRegSRNet	u^2-MDN	Ours
well-registration	PSNR	38.5277	37.2311	42.9390	37.4648	48.2342	38.3979	40.0635	45.4005	**52.2147**
	CC	0.9605	0.9296	0.9812	0.9520	0.9915	0.9625	0.9775	0.9875	**0.9922**
	SAM	5.0334	5.6264	2.9455	4.0079	1.8769	3.6439	2.4781	2.6754	**1.0907**
	ERGAS	3.1643	4.3269	1.9383	3.4866	1.2592	3.2101	2.6118	1.7634	**1.0539**
	RMSE	0.0119	0.0153	0.0072	0.0134	0.0039	0.0120	0.0099	0.0053	**0.0025**
	SSIM	0.9522	0.9483	0.9880	0.9683	0.9944	0.9559	0.9810	0.9924	**0.9976**
α = 200	PSNR	38.4165	36.1447	42.7847	37.1801	48.1760	37.1272	39.8370	46.1824	**51.0830**
	CC	0.9606	0.9190	0.9817	0.9495	0.9895	0.9446	0.9765	0.9867	**0.9908**
	SAM	5.1236	6.2411	2.9954	4.1610	1.7732	4.2895	3.2500	2.1624	**1.2123**
	ERGAS	3.2212	4.7345	1.9582	3.5990	1.3377	3.9267	2.6539	1.6177	**1.1636**
	RMSE	0.0122	0.0167	0.0073	0.0139	0.0039	0.0139	0.0102	0.0050	**0.0028**
	SSIM	0.9524	0.9395	0.9878	0.9654	0.9948	0.9407	0.9763	0.9941	**0.9972**
α = 600	PSNR	35.6728	36.6587	41.4293	34.2103	47.4777	34.9294	38.8332	46.3037	**50.8286**
	CC	0.9192	0.9320	0.9742	0.8935	0.9890	0.9040	0.9651	0.9878	**0.9906**
	SAM	6.5387	5.8414	3.7067	5.6892	1.9963	5.4696	4.2053	2.1975	**1.2207**
	ERGAS	4.4689	4.3950	2.3673	5.0801	1.3954	5.0017	2.9862	1.6066	**1.1918**
	RMSE	0.0171	0.0158	0.0088	0.0199	0.0042	0.0182	0.0114	0.0049	**0.0029**
	SSIM	0.9177	0.9423	0.9829	0.9314	0.9936	0.9028	0.9649	0.9941	**0.9971**
α = 1000	PSNR	33.2087	34.3724	39.1983	32.1783	47.3813	33.5090	38.4490	45.8852	**49.7615**
	CC	0.8608	0.9203	0.9609	0.8256	0.9896	0.8556	0.9661	0.9874	**0.9902**
	SAM	7.9574	7.5768	4.7891	7.0761	1.9712	6.4773	4.0299	2.2289	**1.3540**
	ERGAS	5.8916	5.5402	3.0464	6.3943	1.4144	5.8911	3.0581	1.6959	**1.2379**
	RMSE	0.0224	0.0205	0.0115	0.0250	0.0043	0.0218	0.0120	0.0051	**0.0033**
	SSIM	0.8752	0.9126	0.9723	0.8973	0.9937	0.8663	0.9661	0.9936	**0.9967**

TABLE III

QUANTITATIVE COMPARISON ON CHIKUSEI DATASET WITH VARYING DEGREES OF ELASTIC DEFORMATION. (COLOR CONVENTION: **BEST** AND SECOND-BEST)

Method		CNMF	HySure	GSA	MTF-GLP	MoG-DCN	Integrated	NonRegSRNet	u^2-MDN	Ours
well-registration	PSNR	51.6723	46.3924	48.8327	48.1646	50.0744	46.7507	45.6759	40.6876	**52.5766**
	CC	0.9898	0.9733	0.9890	0.9889	0.9887	0.9881	0.9856	0.9759	**0.9901**
	SAM	1.3040	2.4320	1.6989	1.7485	1.5161	1.6200	1.7209	3.8332	**1.2418**
	ERGAS	2.3263	3.7870	**2.1446**	2.2682	2.6353	2.3391	2.7558	4.8228	2.3103
	RMSE	0.0027	0.0049	0.0037	0.0039	0.0032	0.0047	0.0052	0.0092	**0.0024**
	SSIM	0.9961	0.9904	0.9937	0.9935	0.9942	0.9904	0.9935	0.9685	**0.9970**
α = 200	PSNR	48.2315	45.3000	47.1194	45.1209	49.8846	43.2964	41.8504	40.2546	**52.6546**
	CC	0.9874	0.9711	0.9881	0.9835	0.9888	0.9803	0.9836	0.9789	**0.9903**
	SAM	1.6693	2.6313	1.9445	1.8395	1.5643	2.1590	1.9066	4.2667	**1.2263**
	ERGAS	2.5156	3.9844	2.4263	2.9052	2.4699	3.1128	3.3242	3.7802	**2.2764**
	RMSE	0.0040	0.0056	0.0045	0.0057	0.0033	0.0070	0.0081	0.0097	**0.0024**
	SSIM	0.9933	0.9882	0.9917	0.9893	0.9944	0.9806	0.9892	0.9741	**0.9970**
α = 600	PSNR	38.8806	42.6369	46.5376	38.4183	49.6377	40.5461	40.0379	40.7333	**51.8270**
	CC	0.8865	0.9605	0.9834	0.9353	0.9876	0.9638	0.9729	0.9787	**0.9894**
	SAM	3.9845	3.0168	2.1388	3.0599	1.6320	2.7425	2.7589	4.0390	**1.3277**
	ERGAS	7.7469	4.6056	2.6039	5.6076	2.5994	4.1167	4.2418	3.2644	**2.3663**
	RMSE	0.0148	0.0076	0.0048	0.0123	0.0033	0.0096	0.0099	0.0092	**0.0026**
	SSIM	0.9348	0.9774	0.9905	0.9628	0.9942	0.9656	0.9763	0.9797	**0.9967**
α = 1000	PSNR	31.6077	33.2207	35.9321	30.7469	48.4687	32.6022	39.5230	40.4132	**51.0841**
	CC	0.7169	0.9001	0.9443	0.6483	0.9851	0.7486	0.9674	0.9817	**0.9885**
	SAM	6.3876	7.4070	6.7846	6.8038	1.8766	6.3152	3.1612	4.1215	**1.3853**
	ERGAS	13.1877	9.2949	7.4816	12.3879	2.8494	10.6854	5.0110	3.2183	**2.4156**
	RMSE	0.0269	0.0229	0.0467	0.0297	0.0038	0.0239	0.0166	0.0095	**0.0028**
	SSIM	0.8636	0.8528	0.9201	0.8599	0.9936	0.8461	0.9705	0.9782	**0.9965**

methods, where Integrated, NonRegSRNet, and u^{2}MDN are HSI fusion methods for unregistered images.

1) Results on Elastic Deformation: We perform a comprehensive evaluation of the fusion results of the proposed and comparison methods on Houston and Chikusei datasets. Tables II and III show the results of the objective evaluation. Fig. 6 shows residual maps and corresponding fusion images. The residual maps are calculated based on MSE and the display range is set to 0–0.1, where darker colors and fewer residuals imply higher fusion quality.

The performance of GSA, GLP, HySure, CNMF, and Integrated all decrease as the degree of deformation increases. NonRegSRNet shows stability under different deformation conditions but has not yet achieved satisfactory results. MoG-DCN is based on the model-guided concept and obtains fusion results iteratively, u^2-MDN characterizes the spatial

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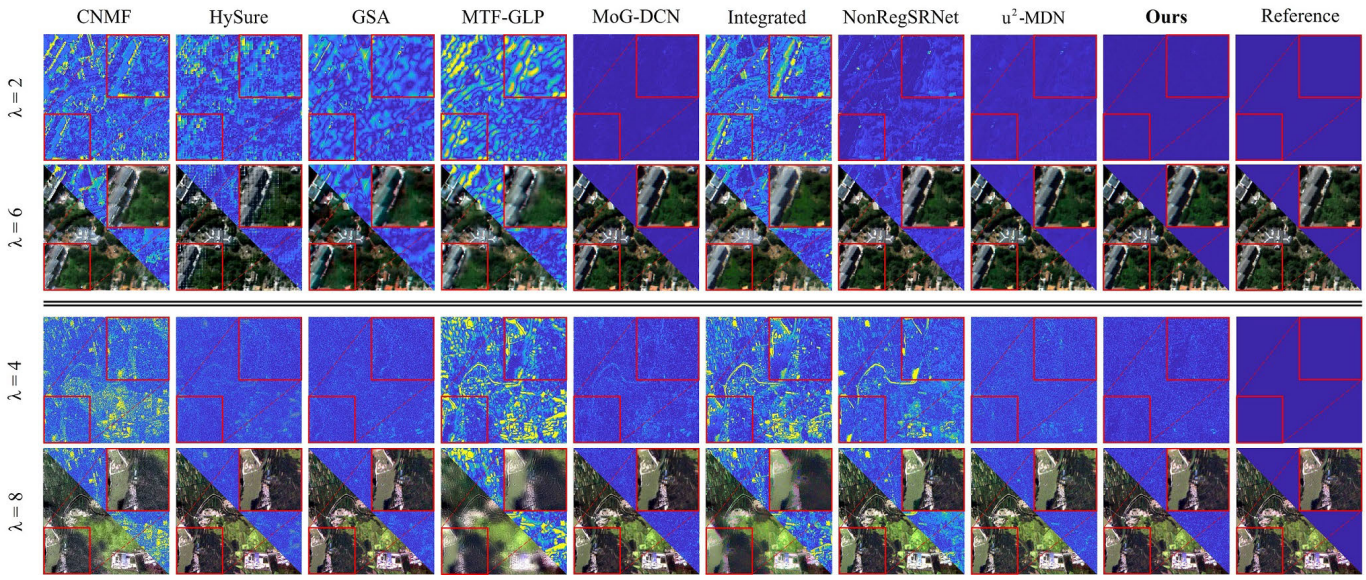


Fig. 7. Visualization results of different methods on Pavia Center and WHU-OHS datasets with different degrees of turbulence deformation.

TABLE IV

QUANTITATIVE COMPARISON ON PAVIA CENTER DATASET WITH VARYING DEGREES OF TURBULENCE DEFORMATION. (COLOR CONVENTION: **BEST** AND SECOND-BEST)

Method		CNMF	HySure	GSA	MTF-GLP	MoG-DCN	Integrated	NonRegSRNet	u^2-MDN	Ours
λ = 2	PSNR	27.8849	28.6367	30.5876	26.3441	42.0878	27.6508	37.4894	41.0009	**42.8527**
	CC	0.8904	0.9332	0.9594	0.8437	0.9889	0.8879	0.9826	0.9884	**0.9903**
	SAM	6.9754	12.0666	10.5188	7.7948	3.1123	6.4929	4.1863	3.7668	**2.9073**
	ERGAS	6.5971	6.1626	4.6687	7.4929	1.8169	6.6198	3.6250	2.6293	**1.6310**
	RMSE	0.0410	0.0373	0.0297	0.0483	0.0079	0.0417	0.0133	0.0089	**0.0072**
	SSIM	0.8321	0.8584	0.9099	0.8218	0.9823	0.8084	0.9722	0.9778	**0.9850**
λ = 4	PSNR	27.7887	29.9491	31.7330	26.3666	41.9768	26.7799	35.5287	40.9515	**42.7681**
	CC	0.8883	0.9456	0.9618	0.8414	0.9888	0.8675	0.9849	0.9879	**0.9901**
	SAM	7.0963	10.6209	8.7909	7.4892	3.0882	6.8806	4.8010	3.7881	**2.9542**
	ERGAS	6.6113	5.3806	4.0954	7.4915	1.8225	7.2607	3.0541	2.5344	**1.6520**
	RMSE	0.0413	0.0322	0.0261	0.0482	0.0081	0.0461	0.0167	0.0089	**0.0073**
	SSIM	0.8279	0.8823	0.9305	0.8214	0.9821	0.7699	0.9659	0.9787	**0.9842**
λ = 6	PSNR	27.3758	29.7806	32.6446	26.1848	41.9955	26.0799	34.4885	40.9200	**42.7275**
	CC	0.8783	0.9328	0.9646	0.8354	0.9886	0.8477	0.9797	0.9887	**0.9900**
	SAM	7.1589	10.0208	7.7683	7.4664	3.1373	7.2459	5.6443	3.7512	**2.9672**
	ERGAS	6.9627	5.5748	3.7347	7.6603	1.8392	7.8405	4.1291	2.6262	**1.6622**
	RMSE	0.0434	0.0328	0.0235	0.0492	0.0081	0.0499	0.0189	0.0089	**0.0073**
	SSIM	0.8127	0.8711	0.9422	0.8106	0.9819	0.7331	0.9532	0.9778	**0.9846**
λ = 8	PSNR	26.5487	27.8088	33.2824	25.8853	41.9972	25.4702	32.1456	40.7468	**43.0758**
	CC	0.8531	0.8469	0.9668	0.8266	0.9890	0.8258	0.9749	0.9882	**0.9906**
	SAM	7.4552	11.4331	7.1922	7.6083	3.2755	7.5875	6.5423	4.0977	**2.8661**
	ERGAS	7.6614	7.7472	3.5364	7.9355	1.8230	8.3968	4.5745	2.5837	**1.6060**
	RMSE	0.0476	0.0416	0.0218	0.0509	0.0081	0.0535	0.0247	0.0092	**0.0070**
	SSIM	0.7737	0.8179	0.9483	0.7933	0.9822	0.6966	0.9349	0.9756	**0.9850**

correlation between different modes by maximizing the mutual information, they can achieve superior fusion results, but there is still some gap compared to the algorithm in this article. Overall, the proposed PMI-RFCoNet proves its effectiveness by obtaining the best visualized images and the darkest residual maps on two datasets with different deformation degrees.

2) Results on Turbulence Deformation: We evaluate the fusion results of the proposed and compared methods on Pavia Center and WHU-OHS datasets with different degrees of turbulence deformation. The results are shown in Fig. 7, Tables IV and V. Similar to the elastic deformation results, traditional methods and Integrated still perform poorly for a wide range of deformation. NonRegSRNet shows better robustness under different deformation levels, but the fusion results are still disappointing. MoG-DCN and u^2-MDN still perform better, but there are some significant highlights in local regions of the residual maps. Overall, our method can maintain the most stable and excellent performance when the image undergoes turbulence deformation.

3) Results on Real-World Dataset: The subjective and objective results of different methods on the XijingU dataset dataset are shown in Fig. 8 and Table VI, respectively. Compared with other methods, the proposed PMI-RFCoNet can obtain HR-HSI with more spatial clarity and more realistic

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TABLE V

QUANTITATIVE COMPARISON ON WHU-OHS DATASET WITH VARYING DEGREES OF TURBULENCE DEFORMATION. (COLOR CONVENTION: **BEST** AND SECOND-BEST)

Method		CNMF	HySure	GSA	MTF-GLP	MoG-DCN	Integrated	NonRegSRNet	u^2-MDN	Ours
$\lambda = 2$	PSNR	26.8694	30.081	32.0535	25.1343	32.2521	26.7148	30.731	31.1653	**34.144**
	CC	0.9036	0.9447	0.9514	0.8597	0.9668	0.896	0.9467	0.9529	**0.9682**
	SAM	4.8339	4.4878	4.374	5.2212	3.664	4.8815	4.4474	4.264	**3.6226**
	ERGAS	3.224	2.4039	2.3002	3.8391	1.8767	3.3331	2.6278	2.2357	**1.8254**
	RMSE	0.0492	0.0359	0.0341	0.0593	0.0278	0.0515	0.0402	0.0338	**0.0271**
	SSIM	0.7114	0.7735	0.7986	0.7272	0.8413	0.7108	0.7825	0.7975	**0.8445**
$\lambda = 4$	PSNR	26.2384	30.1156	31.5083	24.4774	31.9053	25.7966	29.8404	31.146	**34.088**
	CC	0.8914	0.9454	0.9493	0.8376	0.965	0.8774	0.9412	0.9508	**0.9682**
	SAM	5.0736	4.4551	4.4568	5.4845	3.7709	5.1872	4.7607	4.2561	**3.6091**
	ERGAS	3.4543	3.3855	2.3574	4.1283	1.9627	3.6675	2.9676	2.2405	**1.8234**
	RMSE	0.0528	0.0357	0.0351	0.0639	0.0291	0.0566	0.0452	0.0335	**0.027**
	SSIM	0.6929	0.7755	0.7941	0.7104	0.8352	0.6853	0.7642	0.7976	**0.8445**
$\lambda = 6$	PSNR	25.6122	29.9826	31.1545	23.9578	31.5923	25.1750	28.9213	31.2786	**34.0636**
	CC	0.8764	0.9449	0.9477	0.8170	0.9630	0.8587	0.9342	0.9501	**0.9672**
	SAM	5.3073	4.4671	4.496	5.7067	3.859	5.4136	5.0044	4.2309	**3.6643**
	ERGAS	3.7103	2.4020	2.3995	4.3773	1.9982	3.9052	3.2736	2.2363	**1.8435**
	RMSE	0.0568	0.0359	0.0358	0.0678	0.0296	0.0604	0.0499	0.0333	**0.0274**
	SSIM	0.6755	0.7745	0.7923	0.6970	0.8288	0.6674	0.7445	0.7976	**0.8417**
$\lambda = 8$	PSNR	25.292	29.7885	30.8116	23.5311	31.2246	24.5742	28.3778	31.3382	**34.0031**
	CC	0.8684	0.9440	0.9462	0.7974	0.9620	0.8388	0.9273	0.9536	**0.9671**
	SAM	5.4895	4.5065	4.5400	5.9160	3.9266	5.6865	5.1877	4.2118	**3.6635**
	ERGAS	3.8497	2.4383	2.4445	4.5961	2.0416	4.1951	3.4996	2.2298	**1.8504**
	RMSE	0.0589	0.0365	0.0365	0.0713	0.0303	0.0648	0.0535	0.0336	**0.0274**
	SSIM	0.6695	0.7718	0.7898	0.6851	0.8240	0.6493	0.7304	0.8007	**0.8413**

TABLE VI

QUANTITATIVE COMPARISON ON THE REAL-WORLD DATASET. (COLOR CONVENTION: **BEST** AND SECOND-BEST)

Method	CNMF	HySure	GSA	MTF-GLP	MoG-DCN	Integrated	NonRegSRNet	u^2-MDN	Ours
QNR	0.2083	0.4874	0.5656	0.2591	0.5599	0.2041	0.7418	0.6905	**0.8426**
D_λ	0.0836	0.104	0.1272	0.0603	0.0631	**0.0198**	0.0943	0.1271	0.1098
D_s	0.7682	0.4547	0.3543	0.7185	0.4027	0.7919	0.1820	0.2103	**0.0535**

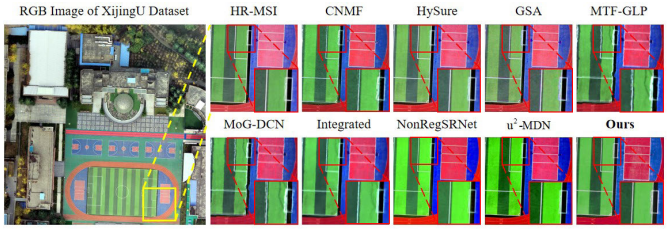


Fig. 8. Visualization results of different methods on the XijingU dataset.

colors on the real-world dataset. In addition, PMI-RFCoNet has the highest QNR value and performs well in spatial and spectral fidelity, which proves its effectiveness in real-world applications.

D. Ablation Studies

In this section, we perform ablation studies on the Houston ($\alpha = 600$), Chikusei ($\alpha = 600$), Pavia Center ($\lambda = 6$), and WHU-OHS ($\lambda = 6$) datasets.

1) Multilevel Registration-Fusion Co-Optimization: In the proposed PMI-RFCoNet, we integrate registration and fusion into the Co-RFB and design it to optimize iteratively on multiple levels. To validate the effectiveness of this multilevel iteration, we perform an ablation study by treating the structure with single-level iteration as a variant of PMI-RFCoNet and naming it PMI-RFCoNet-S. For PMI-RFCoNet-S, we do not perform multilevel registration-fusion co-optimization, only extract features once for the input LR-HSI and HR-MSI before inputting them to Co-RFB. We ensure that the other settings remain unchanged and the parameters are modulated optimally. The results are shown in Fig. 9. Overall, PMI-RFCoNet achieves better performance on all datasets relative to PMI-RFCoNet-S. This is because when LR-HSI undergoes extensive deformation, the multilevel iterative method can not only maintain global consistency, but also accurately adjust its local details, thus proving its effectiveness.

2) Interactive Registration Module: We utilize the deformation field learned by IRM to perform spatial transformation on unregistered image. IRM can continuously focus on the differences and correlations between images when generating deformation field. To demonstrate the effectiveness of this structure, we design a variant of IRM called PMI-RFCoNet-I, which removes the part of information interaction in IRM and directly generates the deformation field by connecting the input image features along the channel dimension. The experimental results on four datasets are shown in Fig. 10. It can be visualized that PMI-RFCoNet obtains better performance than its variant PMI-RFCoNet-I in all quantitative metrics, which proves that IRM is effective in facilitating the generation of more refined deformation field.

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TABLE VII

FUSION RESULTS OF PMI-RFCONET AND ITS VARIANTS THAT BASED ON SRFU ON HOUSTON AND PAVIA CENTER DATASETS

Houston						
Variant	PSNR	CC	SAM	ERGAS	RMSE	SSIM
(a)	45.5263	0.9871	1.7437	1.6329	0.0054	0.9918
(b)	48.3799	0.99	1.6229	1.4468	0.0038	0.996
(c)	49.6868	0.9902	1.4722	1.2775	0.0032	0.9969
(d)	50.3343	0.9894	1.3469	1.2279	0.003	0.9968
(e)	**50.8286**	**0.9906**	**1.2207**	**1.1918**	**0.0029**	**0.9971**
(f)	48.776	0.9893	1.6416	1.3466	0.0036	0.9964
Pavia Center						
(a)	37.3786	0.9827	4.2505	2.8025	0.0138	0.9699
(b)	40.7899	0.9884	3.5443	1.9248	0.0092	0.9807
(c)	42.0019	0.9895	3.1602	1.7854	0.008	0.9825
(d)	41.6989	0.9893	3.2128	1.7719	0.0083	0.9825
(e)	**42.7275**	**0.9900**	**2.9672**	**1.6622**	**0.0073**	**0.9846**
(f)	41.14	0.9892	3.3176	1.8562	0.0088	0.9818

TABLE VIII

MODEL SPACE COMPLEXITY AND TIME COMPLEXITY OF DIFFERENT METHODS

Method	MoG-DCN	NonRegSRNet	u^2-MDN	Ours
Params(M)	7.21	2.13	0.01	13.01
flops(G)	445.28	9.26	1.48	51.69
Time(s)	0.042	0.019	0.71	0.036

3) Spectral Recalibration and Fusion Module: The spectral information in HSIs often exhibits strong correlations, and pixel offsets introduced after registration may lead to spectral deviations. The SRFU proposed in this article is a dual-branch structure that simultaneously ensures the joint enhancement of spectral correction and information fusion. In order to verify the effectiveness of this structure, we design multiple variants based on SRFU: (a) without spatial information fusion branch (w/o B-SpaFus), (b) without spectral correction branch (w/o B-SpeRec), and with different numbers of FC layers [(c) w/FC-0, (d) w/FC-2, (e) w/FC-4, and (f) w/FC-8]. As shown in Table VII, (a) and (b) perform poorly due to the lack of spatial information injection and spectral correction, respectively. In addition, fewer fully connected layers will cause the model adequately capture the relationship between spectral feature channels, while too many layers will be prone to overfitting and lead to degradation of model performance. The experiments show that the model can achieve optimal performance by introducing two branches and setting the number of FC layers to 4 when designing SRFU, which is sufficient to achieve better spectral correction and spatial information enhancement.

E. Discussion

We train all deep learning-based models until they fully converge to the optimum on Pavia Center dataset, using parameters, FLOPS and inference time as measures of space complexity and time complexity. As shown in Table VIII, compared with other methods, our proposed PMI-RFCoNet has the best performance and shorter inference time, but it has a larger number of parameters, which is mainly due to the self-attention operation involved in mining spatial correlations and the fully connected layers used in exploring the dependencies among spectral features. Overall, the proposed

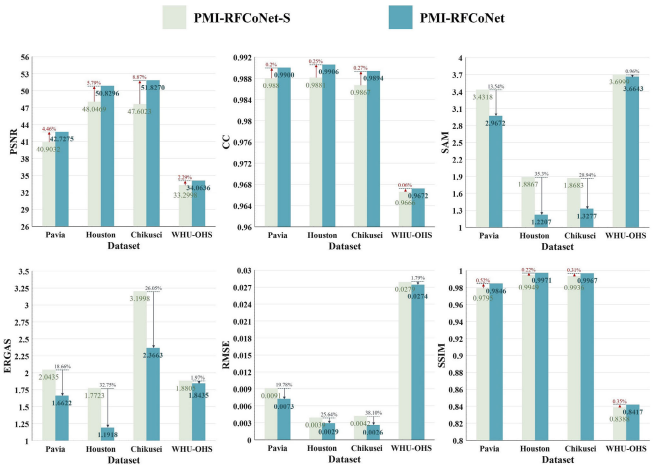


Fig. 9. Fusion results of PMI-RFCoNet and its variant with single-level iteration (PMI-RFCoNet-S) on the Pavia Center, Houston, Chikusei, and WHU-OHS datasets.

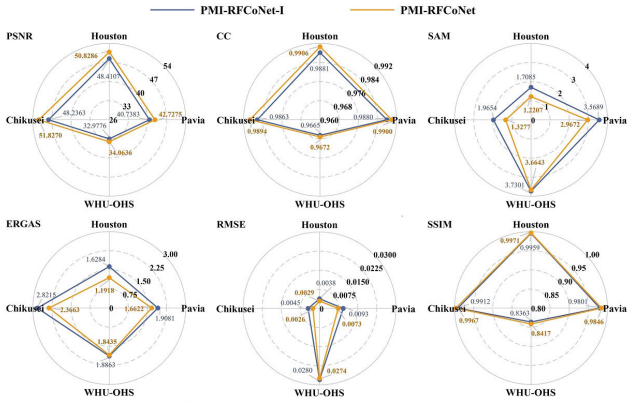


Fig. 10. Fusion results of PMI-RFCoNet and its variant with the removed information interaction structure (PMI-RFCoNet-I) on the Pavia Center, Houston, Chikusei, and WHU-OHS datasets.

PMI-RFCoNet achieves a trade-off between performance and model efficiency. It performs well and shows strong robustness with acceptable space and time complexity. In future research, we will consider compressing the model size while ensuring the high accuracy of HSI-SR.

In addition, although the proposed PMI-RFCoNet can cope with different degrees of non-rigid deformation well, the images may have other unknown degradation types in real scenes. Most of the existing SR methods use images with Gaussian blurring and bicubic downsampling as training samples or use different blurring kernels to simulate the actual scenes, but they still have domain gaps from the real images, which means that when faced images with degradation types that have not been trained on, the use of the existing models may lead to poor performance. Therefore, our future work will introduce the degradation-type estimation in the registration-fusion co-optimization model for better application to real-world scenarios.

V. CONCLUSION

In this article, we propose the PMI-RFCoNet which utilizes the unregistered LR-HSI and HR-MSI to achieve HSI-SR. Co-RFB is the core part of PMI-RFCoNet, and its iteration

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enable the network to fully utilize the spatial and spectral features at the corresponding layers to refine the registration and fusion results step by step. Specifically, the two basic components of Co-RFB are the IRM and the SRFU. The IRM focuses on the correlation between features globally based on information interaction, and generates a finer deformation field to repair non-rigid pixel offsets. The SRFU performs spectral correlation embedding and spatial information injection alternately on the two branches to achieve spectral correction and detail enhancement of the registration results. The experimental results show that PMI-RFCoNet is able to flexibly cope with different degrees of non-rigid deformations and generate high-quality HR-HSI, which fully demonstrates its effectiveness.

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